

TamagotchU!

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Figure 1: Multi-player TamagotchU Installation



Figure 2: fish hands sketch and user emotion analysis test

ABSTRACT

TamagotchU is an emotionally intelligent organism in an immersive XR ecosystem that fosters human dialogue. Running on low-energy edge devices, it critiques large-scale ecological costs through decentralized companionship. Portable tamapagers capture real-time facial and voice emotions, collectively evolving digital pufferfish inspired by Yangtze River indicators. The work proposes eco-sentient systems emphasizing modest computation, collective care, and symbiotic futures over extractive spectacle.

Index Terms: Decentralized ecosystem of AI [frugal computing]; Environmental Storytelling—; Ecological Art [anti monopolistic control of AI development]: —

1 INTRODUCTION

This project reimagines emotionally responsive companions, drawing from the artist's memory of Yangtze River pufferfish, a species signaling environmental health amid data center threats. Participants engage via tama-pagers in immersive projections and soundscapes, where group emotions drive the evolution of TamagotchU fish. Beyond technical description, it explores philosophical questions: How can systems embody ecological interdependence? What playful, frugal models replace energy-intensive systems? TamagotchU prototypes symbiotic coexistence rooted in sensitivity and shared vulnerability.

2 ARTISTIC CONCEPT & MOTIVATION

- The project began with artist's personal memory: growing up in Yangzhong, a small island city on the Yangtze River where puffer fish are everywhere in the rivers and immortalized as giant sculptures facing the wetland frontiers. The puffer fish is more than local folklore, it's a bio-indicator species, surviving

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only when the water condition stays clean and balanced. As vast AI data centers rise near Shanghai and Nanjing, they generate immense heat and resource demand, threatening the river ecosystems these creatures depend on. [1] In TamagotchU, the puffer fish becomes a digital spirit a symbol of fragility, adaptation, and ecological interdependence. By running on modest, distributed edge devices, the work mirrors the ecological resilience it represents: small nodes forming a living network instead of one voracious cloud. The installation merges immersive projection, sound, and collective participation. The audience steps into a space where walls shimmer with responsive visuals, the AI creatures breath beside them as living souls, and every shared emotion becomes a new current in a hybrid ecosystem. Emotion becomes frugal energy, care becomes playful computation. [2]

3 PARTICIPANT EXPERIENCE

Every player holds one of our tama-pager devices built from a single board chip with a camera and microphone, which analyzes their facial expressions and voice tone in realtime through customized lightweight multi modal emotion model. These collective emotional signals ignite TamagotchU fish, a group of AI beings occupying real and virtual space, and each has its own characters and personalities. As participants interact, those creatures evolve, reflecting their shared mood and forming new behaviors. Certain group emotion patterns from the audience trigger the creature give birth of a new human head fused fish.

4 SYSTEM & IMPLEMENTATION

TamagotchU! is built around a multimodal, edge-assisted pipeline that allows audiences to *collectively* shape the emotional state and behavior of a virtual creature in real time. The system integrates three primary input modalities—facial expression, speech, and typed text—which are processed on lightweight edge devices before being aggregated by a central Unity-based server.



Figure 3: player using Tama-pager device interact with fish

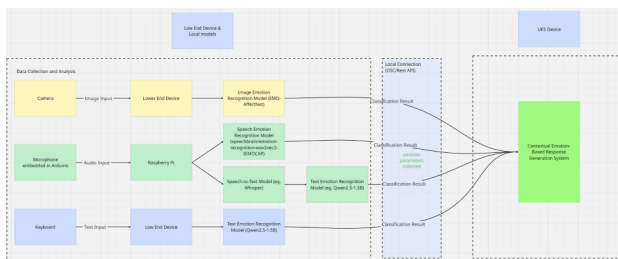


Figure 4: First prototype system architecture.

For facial expression analysis, a webcam continuously captures audience faces at 10 FPS. Frames are processed by a classification model with a ResNet50 backbone [3], which categorizes expressions into seven affective states: Neutral, Happiness, Sadness, Surprise, Fear, Disgust, and Anger. This stream provides a persistent, low-latency signal of the audience's non-verbal affect.

For spoken input, audio is captured via microphone and transcribed using **Whisper-v2-base**, a large-scale weakly supervised speech recognition model optimized for robustness and multilingual performance [4], deployed on a Raspberry Pi using a C++ runtime. The resulting transcripts are further analyzed and mapped to discrete emotion tags. In parallel, participants may also directly type messages via a keyboard interface; these textual inputs are processed using **Qwen3-0.6**, a lightweight large language model optimized for efficient multilingual understanding and reasoning [5], which infers and categorizes the underlying emotional intent.

All sensory inputs—webcam frames, audio-derived transcripts, and typed text—are collected by the edge devices and transmitted to a central server running a Unity application. Within Unity, these aggregated emotional signals dynamically influence TamagotchU's internal affective state, animation parameters, and behavioral policies, enabling a continuously evolving, collectively driven virtual agent.

5 ECOLOGICAL & ETHICAL FRAMING

At a time when AI art is dominated by high performance aesthetics and industrial scale computing, TamagotchU offers a quieter rebellion that celebrates sensitivity over spectacle. Its decentralized, low energy design proposes a model for eco-sentient AI: systems that learn to coexist within the rhythms of both human feeling and environmental boundaries. TamagotchU ultimately asks a simple question: what if our future with AI felt not extractive, but symbiotic?

Rather than feeding massive GPU clusters, it runs locally on our

customized small edge devices, turning AI into a collective companion experience. Conceptually and technologically, TamagotchU responds to the rising energy consumption and environmental cost of large-scale AI systems. It questions the assumption that bigger computation equals better intelligence or deeper art.

6 CONCLUSION

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REFERENCES

- [1] Teaching sustainable creative technologies, 2025. arXiv:2507.05320.
- [2] X. Chen, Z. Ibrahim, and A. A. Aziz. Predicting emotional responses in interactive art using random forests: a model grounded in enactive aesthetics. *Frontiers in Psychology*, 16, 2025. doi: 10.3389/fpsyg.2025.1609103
- [3] K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778, 2016.
- [4] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever. Robust speech recognition via large-scale weak supervision. *arXiv preprint arXiv:2212.04356*, 2022.
- [5] A. Yang et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.